Unified Assembly Hierarchy Architecture

The Digital Thread: Connecting Design, Kitting, Assembly Execution & Timesheet Costing

ARCHITECTURE PLAN
Target: Production MES
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Status: Approved Proposal

01 Operational Reality vs. The Digital Gap

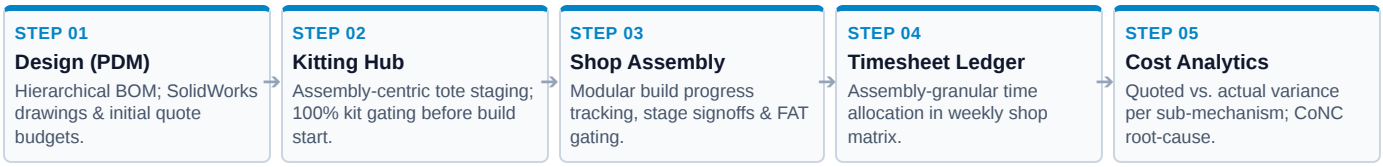
In custom automation machinery manufacturing at VTR, machines are engineered, staged, and assembled as a modular hierarchy of **Assemblies and Sub-Assemblies** (e.g., *Bowl Drive Unit, Linear Track Feed, Escapement Mechanism, Pick-and-Place*).

The Shop Floor Reality Today: The BOM from Design is *already passed to Kitting* so that parts are physically sorted and "kitted" into totes according to assemblies. Similarly, the design drawings passed to the Assembly team are *already organized according to these same assemblies*.

The Software Gap in Flux: The `vtrFlux` application does not have this infrastructure yet:

- **In Kitting:** Flux presents parts as a flat list, lacking the assembly tote containers that clerks physically build.
- **In Timesheets:** Workers holding physical assembly drawing packages are forced to log hours against generic machine buckets.
- **In Quality:** SOP 1-003 rework records sub-assemblies only as unvalidated, free-form text strings.

FIGURE 1: THE UNIFIED DIGITAL THREAD LIFECYCLE (DESIGN → KITTING → BUILD → COSTING) CONTINUOUS DATAFLOW



02 Relational Entity Architecture & Database Schema

Rather than separate rigid tables, Flux will use a **recursive adjacency tree** (`parent_assembly_id`) in PostgreSQL. This allows infinite sub-assembly nesting while enabling fast queries via recursive Common Table Expressions (CTEs).

```
-- 1. Create recursive assemblies table with self-referencing parent
CREATE TABLE machine_assemblies (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    machine_id UUID NOT NULL REFERENCES machines(id) ON DELETE CASCADE,
    parent_assembly_id UUID REFERENCES machine_assemblies(id) ON DELETE CASCADE, -- NULL for Root Node
    assembly_code VARCHAR(50) NOT NULL,        -- e.g. "ASM-01", "SUB-01A"
    name VARCHAR(255) NOT NULL,                -- e.g. "Centrifugal Bowl Drive", "Singulator Mechanism"
    description TEXT,
    drawing_number VARCHAR(100),               -- SolidWorks PDM Drawing Reference
    revision VARCHAR(20) DEFAULT 'A',
    department VARCHAR(50) NOT NULL DEFAULT 'assembly',
    quoted_hours NUMERIC(6, 2) NOT NULL DEFAULT 0.00,
    status VARCHAR(50) NOT NULL DEFAULT 'design_draft',
    sequence_order INT NOT NULL DEFAULT 0,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    CONSTRAINT uq_machine_assembly_code UNIQUE (machine_id, assembly_code)
);

-- 2. Link Timesheets, Kitting Parts, and Rework directly to Assembly Nodes
ALTER TABLE time_entries ADD COLUMN assembly_id UUID REFERENCES machine_assemblies(id) ON DELETE SET NULL;
ALTER TABLE kitting_parts ADD COLUMN assembly_id UUID REFERENCES machine_assemblies(id) ON DELETE CASCADE;
ALTER TABLE rework_items ADD COLUMN assembly_id UUID REFERENCES machine_assemblies(id) ON DELETE SET NULL;

CREATE INDEX idx_assemblies_machine ON machine_assemblies(machine_id);
CREATE INDEX idx_assemblies_parent ON machine_assemblies(parent_assembly_id);
CREATE INDEX idx_time_entries_assembly ON time_entries(assembly_id);
CREATE INDEX idx_kitting_parts_assembly ON kitting_parts(assembly_id);
```

FIGURE 2: UNIFIED ASSEMBLY TREE & DOMAIN LINKAGES POSTGRESQL HIERARCHY MODEL

MACHINE: VTR-1024 Centrifugal Feeder (Order # 1024) MACHINE SCOPE ROOT

ASM-01: Bowl Drive Assembly 14.5H / 16H QUOTED

Drawing: VTR-1024-D01 Rev B • Status: Assembled

- **SUB-01A: Motor & Mount:** 3.5h actual (4h quote) • 2 Parts Kitted
- **SUB-01B: Reduction Gearbox:** 6.0h actual (6h quote) • 4 Parts Kitted
- *Timesheet Attribution:* `time_entries.assembly_id` ➔ ASM-01

ASM-02: Escapement & Track 38H / 24H (OVER BUDGET!)

Drawing: VTR-1024-D02 Rev C • Status: In Assembly

- **SUB-02A: Singulator Gate:** 18.5h actual (8h quote) • Rework Fitment
- **SUB-02B: Pneumatic Slide:** 11.5h actual (10h quote) • 100% Kitted
- *Rework & CoNC Attribution:* `rework_items.assembly_id` ➔ SUB-02A

03 Departmental Workflows & Timesheet Integration

1. CAD Design & CSV Ingestion Provision

Design engineers already group the entire machine build by assembly in CAD. Design will soon export structured CSV BOMs. Flux provides an extensible `BOMImporter` interface and two-stage preview/commit pipeline ready to plug in as soon as the format is finalized.

2. Kitting Staging Gating

Kitting clerks pick parts into assembly-labeled totes. An assembly only transitions to `kitted` when 100% of required parts are verified, preventing technicians waiting on missing parts.

3. Frictionless Timesheet Logging

When selecting a machine in `WeeklyTimesheetGrid`, an optional *Assembly* dropdown cascades. Selecting an assembly is 1-click; general shop or framing hours can leave it blank without friction.

4. FAT Readiness Shield Integration

Uncompleted assemblies act as active blockers on the machine's Dashboard FAT Readiness Shield, guaranteeing that no machine is scheduled for customer runoff until all sub-units pass inspection.

04 Labor Variance & Assembly Costing Analytics

By capturing quoted budgets at the assembly node and actual hours via weekly time entries, Flux computes automated variance and cost overruns per mechanism:

Assembly Node & Sub-Units	Drawing / Rev	Quoted	Actual	Variance	Labor Cost ($85/hr)	Status
ASM-01 Bowl Drive Assembly	VTR-1024-D01 Rev B	16.0h	14.5h	**-1.5h**	$1,232.50	ASSEMBLED
↳ SUB-01A Motor & Mount Plate	VTR-1024-D01-1	4.0h	3.5h	-0.5h	$297.50	ASSEMBLED
↳ SUB-01B Reduction Gearbox	VTR-1024-D01-2	6.0h	6.0h	0.0h	$510.00	ASSEMBLED
ASM-02 Escapement & Track	VTR-1024-D02 Rev C	24.0h	38.0h	**+14.0h ⚠**	**$3,230.00**	IN ASSEMBLY
↳ SUB-02A Singulator Gate Mechanism	VTR-1024-D02-1	8.0h	18.5h	**+10.5h ⚠**	$1,572.50	REWORK FIT
↳ SUB-02B Pneumatic Infeed Track	VTR-1024-D02-2	10.0h	11.5h	+1.5h	$977.50	TESTING
ASM-03 Stand, Guarding & Controls Base	VTR-1024-D03 Rev A	12.0h	11.0h	**-1.0h**	$935.00	ASSEMBLED
TOTAL ASSEMBLY LABOR ROLLUP	**7 Sub-Units**	**52.0h**	**63.5h**	**+11.5h Over**	**$5,397.50**	78% COMPLETE

05 Implementation Roadmap

Following the recently established 3-tier domain architecture, the feature will be engineered in four non-breaking, incremental phases:

Phase 1: Domain Core & Schema — FOUNDATION

- Create `machine_assemblies` table with self-referential parent key.
- Implement `internal/domain/assemblies/` (repository & service).
- Add REST endpoints: `GET/POST/PUT/DELETE /api/machines/{id}/assemblies`.
- Unit & integration tests validating recursive hierarchy serialization.

Phase 2: Timesheet Labor Attribution — SHOP FLOOR ROI

- Add `assembly_id` FK column to `time_entries` table.
- Update `WeeklyTimesheetGrid` & `TimesheetGridRow` with optional assembly selector.
- Update labor correlation queries to compute actual vs. quote per assembly.
- Verify timesheet submission workflows remain 100% backward compatible.

Phase 3: Kitting Staging Engine — PARTS STAGING

- Link `kitting_parts.assembly_id` to target assembly nodes.
- Overhaul Kitting Hub UI into an assembly-centric kit staging station.
- Implement 1-click part pick toggles and assembly completeness gating.
- Broadcast real-time SSE event `assembly_kitted` to assembly technicians.

Phase 4: Assembly Shop Hub & FAT Gating — EXECUTIVE VISIBILITY

- Create dedicated Assembly Hub with stage progress Kanban boards.
- Wire uncompleted assemblies into the Dashboard FAT Readiness Shield.
- Expose assembly-level labor variance metrics to team leads and quoting estimators.
- Connect SOP 1-003 rework items directly to assembly root causes.

06 Architectural Guardrails & Sign-Off Decisions

Guaranteed Backward Compatibility Safeguards:

- **Non-Disruptive Adoption:** Existing machines without defined assembly hierarchies continue operating normally. Top-level machine tasks (e.g. *Mechanical Assembly*) remain valid fallback targets.
- **Nullable Foreign Keys:** `time_entries.assembly_id` is strictly nullable. Non-assembly shop floor tasks (indirect labor, shop cleanup, general troubleshooting) will never be blocked by mandatory assembly validation.
- **Sub-Millisecond Recursive Queries:** Typical machine assembly hierarchies range between 5 and 30 nodes. PostgreSQL recursive CTE queries execute in < 2ms, maintaining instant UI loading on the shop floor.

DECISION GATE SIGN-OFF MATRIX:

1. BOM Ingestion Strategy:	Manual tree builder in Phase 1; provision pluggable BOMImporter interface with CSV preview/commit pipeline once export structure is finalized.
2. Quoting Authority:	Allow quoted hours at both the Assembly level and Task level, rolling up automatically.
3. Timesheet Usability:	Keep Assembly selection optional on weekly matrix; highlight assembly context when active.